

PRANJAL GUPTA

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#)

Skills

- **Programming:** Java, Python, JavaScript
- **Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
- **Web Development:** React.js, Node.js, Express.js, Flask, REST APIs, JWT Authentication, Socket.IO
- **Databases:** MongoDB, MySQL
- **Artificial Intelligence & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, OpenCV, CNNs, Computer Vision
- **Generative AI:** LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases (ChromaDB)
- **Cloud & DevOps:** AWS (EC2, S3), Docker, Git, GitHub
- **Developer Tools:** Postman, VS Code, Jupyter Notebook, OpenProject, Taiga

Languages

Hindi: Native
English: Proficient (C2)
Japanese: Beginner

Summary

Computer Science undergraduate specializing in AI & ML with experience building full-stack web applications, retrieval-augmented generation (RAG) systems, and AI-powered developer tools. Skilled in Java, Python, JavaScript, React, Node.js, LangChain, and backend development, with a strong interest in software engineering and applied AI.

Education

B.Tech CSE (AI & ML), Faculty of Technology, University of Delhi
2023–2027

CGPA: 7.60 / 10.0

Year 1-7, Year 2-7.77, Year 3-8

CBSE Class XII

91% – St. Andrews Public School

2022

CBSE Class X

93% – St. Andrews Public School

2020

Projects

- **Hybrid Audio Classification Pipeline** – Built an audio classification system combining Wav2Vec2 embeddings with MFCC, YIN, and Spectral Rolloff features for biological vs synthetic speech detection using deep learning techniques.
- **Real-Time Multi-Chat Application** – [GitHub](#)
Developed a full-stack chat platform using React, Node.js, MongoDB, and Socket.IO with JWT authentication, refresh-token workflows, private messaging, and real-time communication.
- **Space Debris Position Prediction** – Compared analytical SGP4 orbital prediction with LSTM-based sequence modeling for satellite trajectory forecasting and performance evaluation.
- **RAG-Based Interview Question Generator** – [GitHub](#) [Live](#)
Built a LangChain-based Retrieval-Augmented Generation system that generates interview questions and model answers from uploaded documents using semantic retrieval and vector embeddings.
- **Personal Portfolio with AI Assistant** – [GitHub](#) [Live](#)
Developed a portfolio platform featuring a personalized AI assistant using LangChain, Gemini embeddings, ChromaDB, and FastAPI for document-grounded question answering.

Activities

Hackathons

2025–26

- Participated in team-based software development and rapid prototyping competitions.

Volunteer

Indian DefSpace Symposium 2025

- Supported event coordination and technical session logistics during a national-level symposium.